

MIT 18.06SC - Unit II Exam

1) a) $\det = \pm 1$ because it is a square orthogonal matrix

b)
$$\begin{vmatrix} q_1 + q_2 & q_2 + q_3 & q_3 + q_1 \end{vmatrix}$$

$$= \begin{vmatrix} q_1 & q_2 & q_3 \end{vmatrix} + \begin{vmatrix} q_2 & q_3 & q_1 \end{vmatrix}$$

$$= \pm 2$$

The $\det$ is a linear function of the columns.

c) $\det = 1$ because the ~~second~~ both matrices have $\det \pm 1$ but since they are the same ~~aff~~ matrix after an even number of column exchanges, the determinants have the same sign.

2) a) $Ax = b$:-
$$\begin{bmatrix} 1 & -10 \\ 1 & -9 \\ \vdots & \vdots \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 9 \\ 1 & 10 \end{bmatrix} \begin{bmatrix} c \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

Not solvable so instead we solve $A^T A \hat{x} = A^T b$:-

$$\begin{bmatrix} 21 & 0 \\ 0 & 245 \\ & 170 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\hat{C} = 1/21 \quad \hat{D} = 0$$

$$b = \frac{1}{21}$$

b) We are projecting b onto $C(A)$ and the columns of A are its basis.

$$\begin{bmatrix} 1 & 1 & \dots & 1 & 1 \\ -10 & -9 & \dots & 9 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$b = A \hat{x} + e$$

$$e = b - A \hat{x}$$

$$= \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} - \begin{bmatrix} 1 & -10 \\ 1 & -9 \\ \vdots & \vdots \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 10 \end{bmatrix} \begin{bmatrix} 1/21 \\ 0 \end{bmatrix}$$

$$= [-1/21 \quad -1/21 \quad \dots \quad 20/21 \quad \dots \quad -1/21]^T$$

↓
21 entries, 20/21 in the middle

$$3) a) P_Q = Q (Q^T Q)^{-1} Q^T \\ = Q I_3 Q^T \\ = Q Q^T$$

$$P_A = A (A^T A)^{-1} A^T$$

b) $P_Q \neq P_A$ because A and Q have the same do not necessarily have the same column space.

$$P_Q Q = Q Q^T Q = Q I_3 = Q$$

$$\det P_Q = 0$$

c) Number 3

4) a) Assuming 'any numbers' cannot be X, the largest possible degree is 2, because:-

$$\det(A) = \sum_{24} a_{1\alpha} a_{2\beta} a_{3\gamma} a_{4\omega}$$

According to the formula we can only choose one entry from each row and column of the so $\rightarrow$ for each term, so at most two X can be chosen.

$$\bullet \text{ b) } X(1) + X(-X) + X(-X) + X(-X) \\ X - X^2 - X^2 - X^2 \\ X - 3X^2$$

To get $\det(A) = 0$:-

$$X - 3X^2 = 0 \\ X(1 - 3X) = 0$$

$$\det(A) = 0 \text{ when } X = 0 \text{ or } X = \cancel{\sqrt{1/3}} \cancel{1/3} 1/3$$